HDOC - 68th day

DECIMAL TO BINARY USING RECURSION

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Problem Statement

Write a recursive function to convert a decimal number into its binary representation.

The program should read a decimal integer, recursively divide it by 2, print the remainders in reverse order, and display the equivalent binary representation.

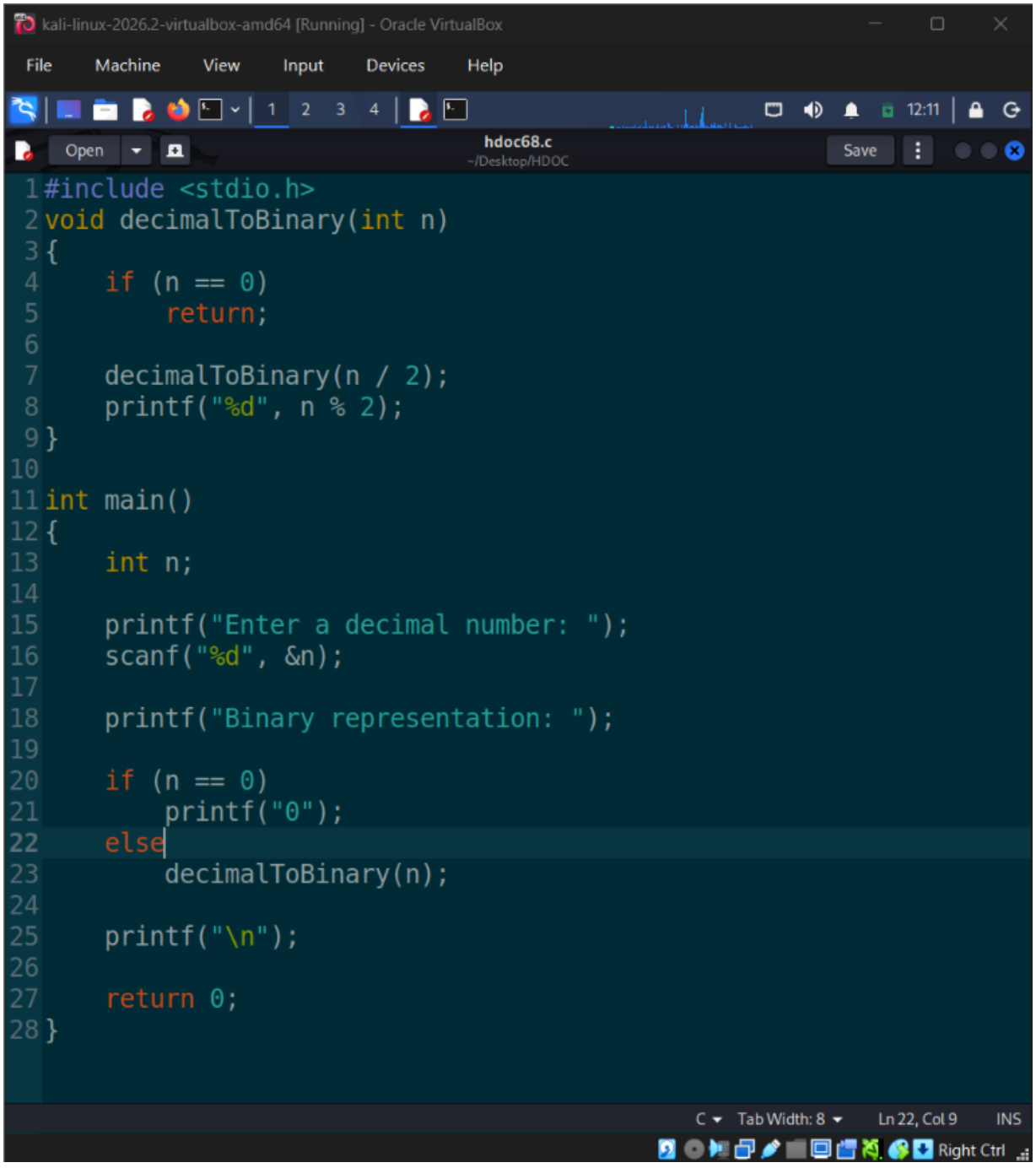
Algorithm

1. Start.
2. Read a decimal integer n.
3. If n = 0, display 0.
4. Otherwise, call the recursive function decimalToBinary(n).
5. In the function, if n = 0, return.
6. Recursively call decimalToBinary(n / 2).
7. Print n % 2 while returning from recursion.
8. Stop.

Test Case Table

S. No.	Decimal Number	Expected Output	Actual Output	Result
1	45	Binary = 101101	Binary = 101101	Pass
2	101	Binary = 1100101	Binary = 1100101	Pass
3	2	Binary = 10	Binary = 10	Pass

C Program




```c
#include <stdio.h>
void decimalToBinary(int n)
{
    if (n == 0)
        return;

    decimalToBinary(n / 2);
    printf("%d", n % 2);
}

int main()
{
    int n;

    printf("Enter a decimal number: ");
    scanf("%d", &n);

    printf("Binary representation: ");

    if (n == 0)
        printf("0");
    else
        decimalToBinary(n);

    printf("\n");

    return 0;
}
```

Compilation and Execution

Compilation command: gcc hdoc68.c -o hdoc68



Execution command: ./hdoc68

Execution Screenshot:

